Interstellar Harvest: Growing Food Beyond Earth

ASSESSMENT RUBRIC · LDTC 615

Module 1 Assignment Rubric

Mission Briefing: Why Your Destination Is the Hardest

Total: 100 points | **CLOs:** 1, 4 | **Length:** 300–400 words | **Type:** Analytic · Weighted criteria

CRITERIA	EXCELLENT 100–90%	PROFICIENT 89–75%	DEVELOPING 74–60%	BEGINNING BELOW 60%
Identification of Environmental Challenges 25% of total	Accurately identifies two or three significant environmental challenges specific to the chosen destination. Each challenge is named and described with enough precision to distinguish it from general statements — explains what low atmospheric pressure does to plant cells, not just that the Moon lacks atmosphere.	Identifies relevant challenges but descriptions lack specificity in one or two places. Challenges are correct but some explanations stay at the surface level without describing the mechanism.	Identifies one or two challenges but descriptions are vague or partially inaccurate. May confuse conditions across destinations or name challenges without connecting them to plant survival.	Challenges identified are incorrect, too vague to demonstrate understanding, or not specific to the chosen destination.
Biological Accuracy: Effect on Plants 30% of total	Clearly explains what each identified condition does to plant biology at a mechanistic level. For example: what cell rupture means and why it happens under low pressure, how radiation damages DNA, or what microgravity does to root orientation and fluid dynamics. Explanations are accurate and specific without requiring the reader to infer the connection.	Explains biological effects for most conditions but one explanation is incomplete or overly general. The connection between condition and plant response is present but not fully developed.	Mentions biological effects but without explaining the mechanism. Stays at cause-effect level without describing what physically or biologically happens inside the plant.	Biological effects are absent, incorrect, or confused. The response describes conditions without connecting them to plant biology in any meaningful way.
Relevance to Long-Duration Spaceflight 25% of total	Clearly articulates why the identified challenges matter for a crew on a long-duration mission to the chosen destination. Goes beyond "food is important" to address what the mission cannot function without and why the specific destination makes it harder than other locations.	Addresses mission relevance but in general terms. Connects food production to the mission without specifying what fails without it or why the specific destination compounds the difficulty.	Mentions mission relevance but the connection is weak or generic — could apply to any destination rather than the one chosen.	No meaningful connection made between the agricultural challenges and mission survival or function.
Unanswered Question 10% of total	Poses a specific, genuine question that the briefing itself does not answer. Points toward a real engineering or scientific challenge the crew would need to solve. Shows awareness that the problem is not fully solved.	Question is relevant but somewhat broad or predictable. Points in the right direction without demonstrating deep engagement with the unsolved challenge.	Question is vague or restates something already covered in the briefing. Does not point toward a real unsolved problem.	No question posed, or question is unrelated to the chosen destination's agricultural challenges.
Writing Clarity & Plain Language 10% of total	Writing is clear, well-organized, and accessible to a non-specialist reader as the prompt instructs. Technical terms are explained when introduced. Stays within the 300–400 word range.	Mostly clear with minor lapses into unexplained jargon or slight organizational issues. Stays close to the word range.	Writing is uneven. Some sections are clear, others assume knowledge the audience does not have. May be significantly over or under the word range.	Writing is unclear or heavily jargon-dependent without explanation. A crew member without botany background could not follow the briefing.

SCORE KEY ● Excellent: 90–100 pts ● Proficient: 75–89 pts ● Developing: 60–74 pts ● Beginning: Below 60 pts

EQUITY & ACCESSIBILITY NOTE

The prompt explicitly invites plain language and does not require scientific vocabulary. A learner who explains what radiation does to DNA in everyday terms scores as well as one who uses technical terminology correctly. The rubric rewards clarity of reasoning over demonstration of prior knowledge. The 300–400 word range and paragraph format requirements are kept minimal to reduce barriers for learners with varying writing backgrounds or language fluency.